Areas under multiple curves

1 For each of the following pairs of functions:

i Graph the functions on the same set of axes.

ii Hence, find the area of the region bounded by the functions and the x-axis.

a $x + y = 2$ and $y = x$.

b $x + y = 6$ and $y = 2x$.

2 For each of the following pairs of functions:

i Graph the functions on the same set of axes.

ii State the value(s) of x at which the two functions intersect.

iii Hence, find the area of the region bounded by the functions and the x-axis.

a $y = 2x$ and $y = (x - 4)^2$.

b $y = x^2$ and $y = (x - 2)^2$

3 Find the area of the region bounded by the following pairs of functions and the x-axis:

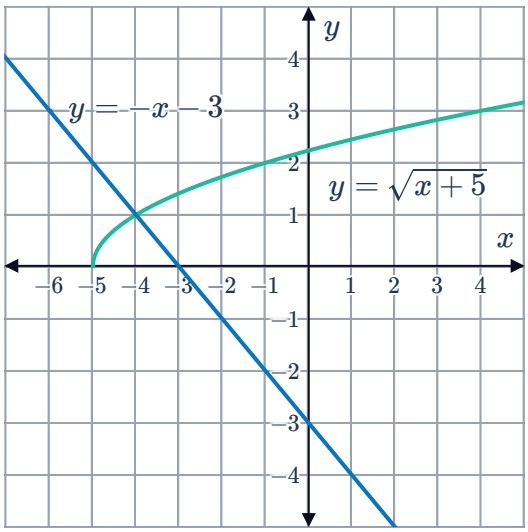
a $y = (x - 1)^2$ and $y = (x + 3)^2$

b $y = x^2 - 4$ and $y = -2x + 11$.

c $y = x^3$ and $y = -3x + 14$.

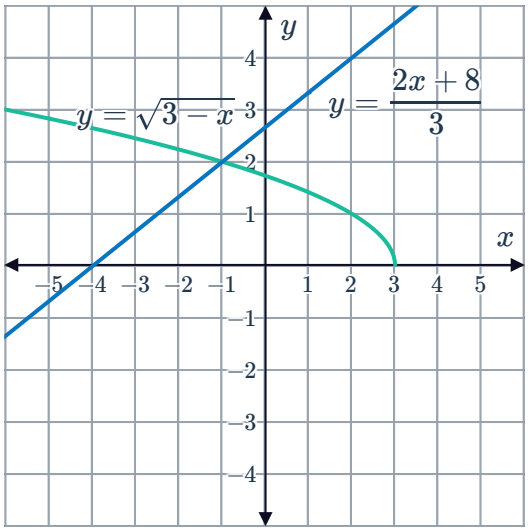
d $y = (x + 2)^2$ and $y = -4x - 3$.

4 Find the area of the region bounded between the curve $y = \sqrt{x + 5}$, the line $y = -x - 3$ and the x-axis:

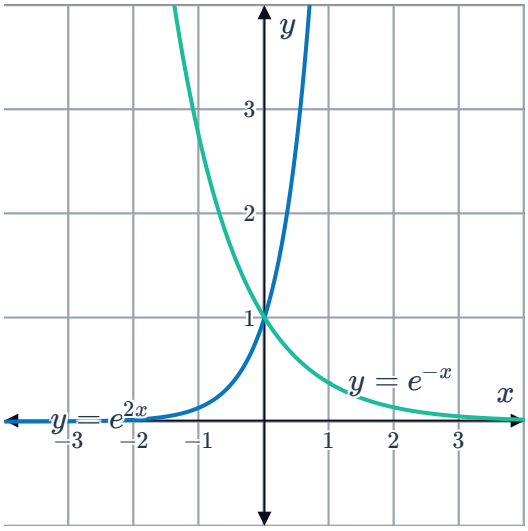


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5 Find the area of the region bounded between the curve $y = \sqrt{3 - x}$, the line $y = \dfrac{2x + 8}{3}$ and the x-axis:

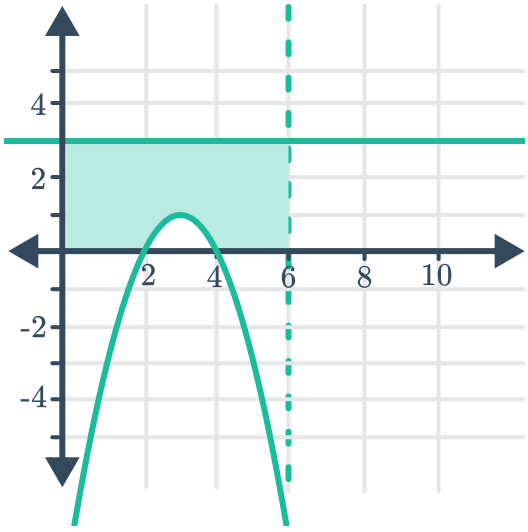


6 Find the area of the region bounded by the curves $y = e^{2x}$ and $y = e^{-x}$, the x-axis, and the lines $x = -2$ and $x = 2$:



7 Find the area of the region bounded by $y = x^2 + 1$, $y = -\dfrac{x}{2} + 6$, the x-axis, and the y-axis.

8 The diagram shows the shaded region bounded by $y = 3$, $y = 0$, $y = 6x - x^2 - 8$, $x = 0$ and $x = 6$:
Find the area of the shaded region.



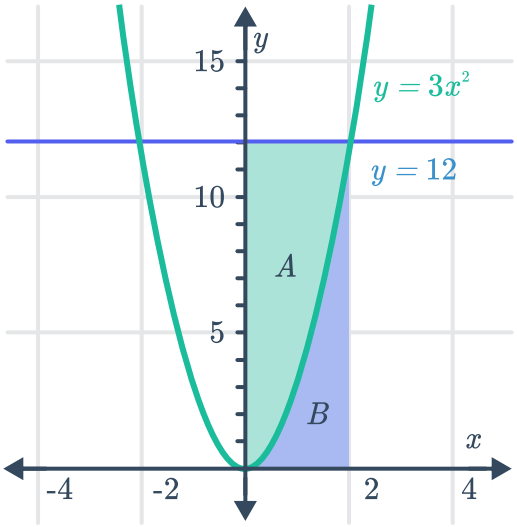
9 Consider the following graph:

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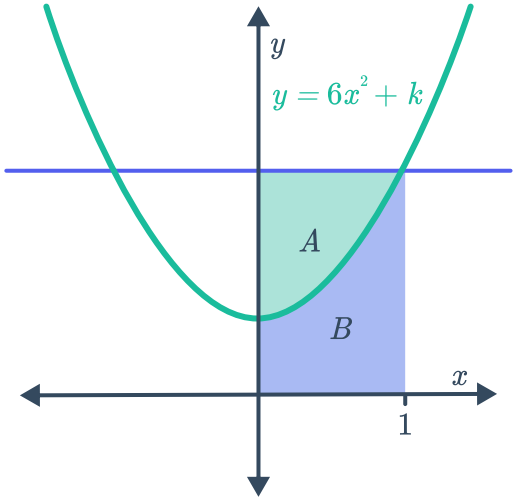
a Find the area of the shaded region B.

b Find the area of the shaded region A.

c What is the ratio of the areas $A : B$?



10 Consider the following graph:
Find k such that the two shaded regions have equal area.



11 Find the exact area between the graph of $y = \sqrt{4 - x^2}$ and the lines $x = 2$ and $y = 2$.

12 Consider the functions $y = e^{\frac{x}{6}}$ and $y = e^{\frac{3}{2}}$.

a Find the x-coordinate of the point of intersection.

b Calculate the exact area bounded by $y = e^{\frac{x}{6}}$, the y-axis and the line $y = e^{\frac{3}{2}}$.

13 Consider the curve $y = e^x$.

a Find the area bound by the curve, the x-axis, the y-axis, and the line $x = 3$.

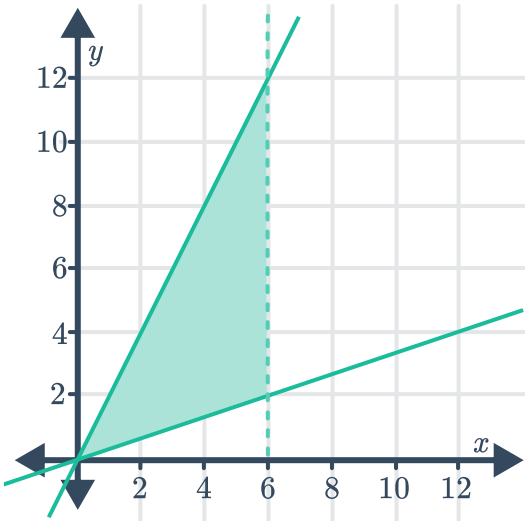
b Find the equation of the tangent to the curve $y = e^x$ at the point where $x = 3$.

c Find the exact area enclosed between $y = e^x$, the x-axis, the y-axis, and the tangent at $x = 3$.

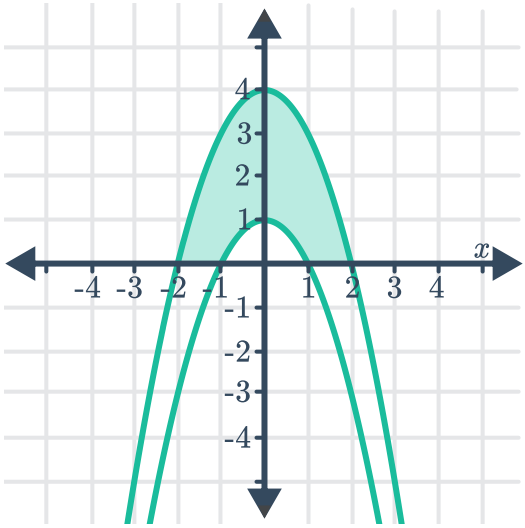
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Areas between curves

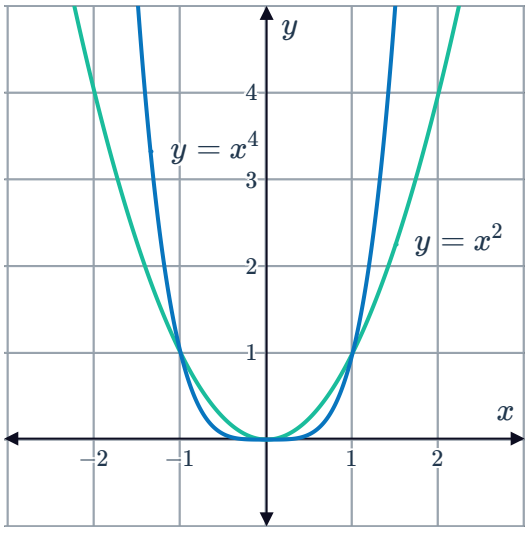
14 Find the area enclosed between the lines $y = 2x$, $y = \frac{1}{3}x$ and $x = 6$:



15 The diagram shows the shaded region bounded by $y = 4 - x^2$, $y = 1 - x^2$ and the x-axis:
Find the area of the shaded region.

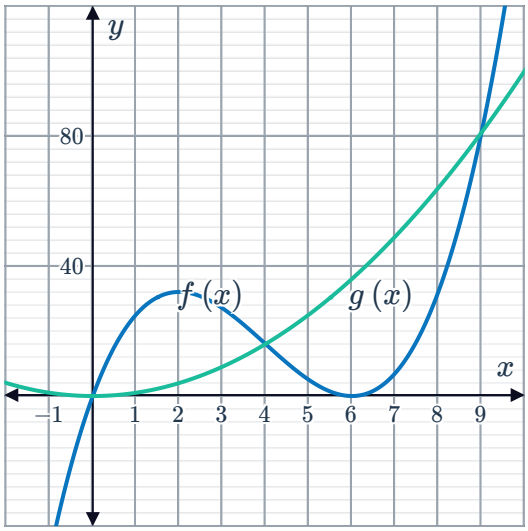


16 Consider the graph of the functions $y = x^2$ and $y = x^4$:
Find the area enclosed between the two curves.



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17 Consider the graph of the functions $f(x) = x(x-6)^2$ and $g(x) = x^2$:



- **a** State the values of x at which the curves intersect.
- **b** Hence, find the total area bounded between the curves.
- **c** Find the area bounded by the curves and the x-axis.

18 For each of the following pairs of functions:

- **i** Graph the functions on the same set of axes.
- **ii** State the values of x at which the line and the curve intersect.
- **iii** Hence, find the area enclosed between the line and the curve.

- **a** $y = x^2$ and $y = x + 2$
- **b** $y = x(x-4)$ and $y = x$
- **c** $y = -x^2 + 8$ and $y = -x + 2$
- **d** $y = x(x-4)^2$ and $y = x$
- **e** $y = -x(x-3)^2$ and $y = -x$

19 For each of the following pairs of functions:

- **i** Find the values of x at which the two curves intersect.
- **ii** Find the area enclosed between the two curves.

- **a** $y = x^2 - 48$ and $y = -(x-2)^2 + 4$
- **b** $y = x$ and $y = (x-5)^3 + 5$
- **c** $y = 4x - 12$ and $y = x(x-3)^2$
- **d** $y = 2x$ and $y = x^2 - 15$

20 Consider the functions $y = -2x(x-4)$ and $y = -x + 4$.

- **a** Graph the functions on the same set of axes.
- **b** State the values of x at which the curve and the line intersect.
- **c** Hence, find the area enclosed between the curves.
- **d** Find the small area enclosed between the curve, the line and the y-axis.

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21 Consider the functions $y = x^2$ and $y = 8 - x^2$.

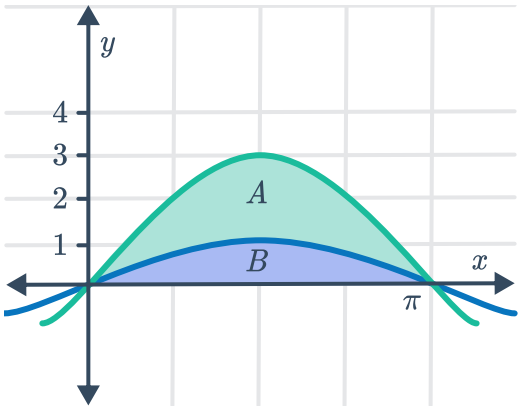
a Graph the functions on the same set of axes.

b State the values of x at which the curves intersect.

c Hence, find the area bounded between the curves.

d Find the small area bounded between the curves and the x-axis, correct to one decimal place.

22 The diagram shows the graphs of $y = \sin x$ and $y = 3\sin x$:



a Determine whether the following integral could be performed to calculate the area of the region bound by the curves, A.

i $\int_0^{\pi} (3\sin x - \sin x)\, dx$

ii $\int_0^{\pi} 3\sin x dx - \int_0^{\pi} \sin x dx$

iii $\int_0^{\pi} 3\sin x dx + \int_0^{\pi} \sin x dx$

iv $2\int_0^{\pi} \sin x dx$

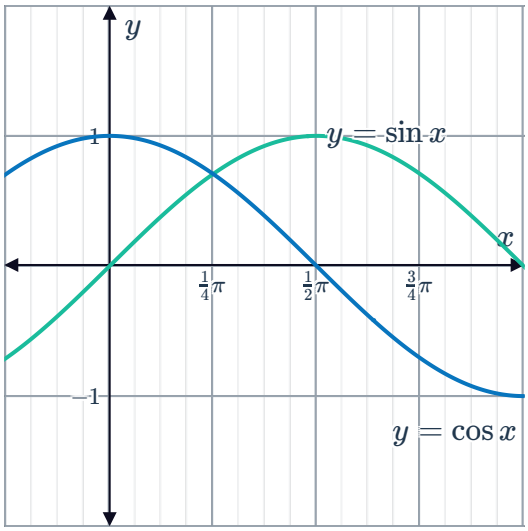
v $\int_0^{\pi} (3\sin x + \sin x)\, dx$

vi $4\int_0^{\pi} \sin x dx$

b Find the area of the shaded region A.

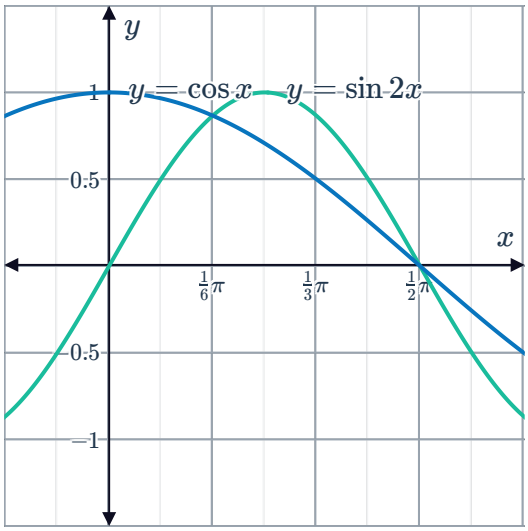
c What is the ratio of the areas $A : B$?

23 Find the area of the region bounded by the curves $y = \sin x$ and $y = \cos x$, and the lines $x = 0$ and $x = \dfrac{\pi}{4}$:

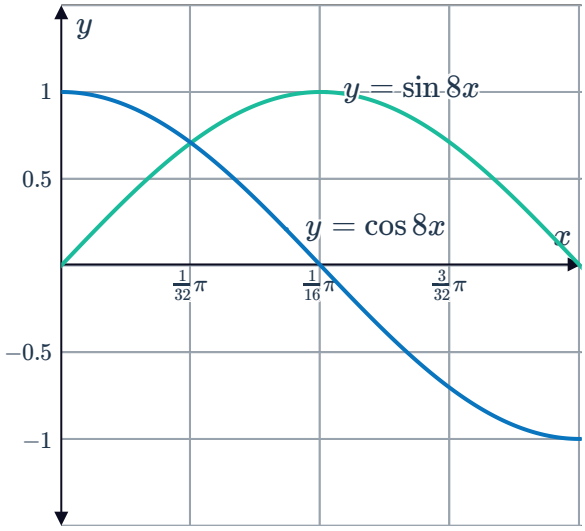


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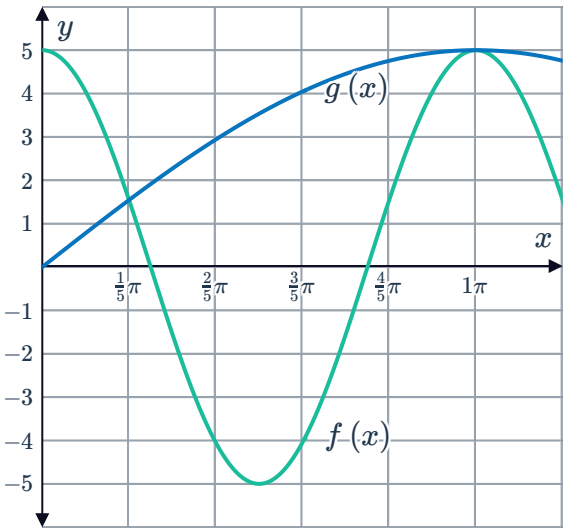
24 Find the area enclosed by the two curves of $y = \sin 2x$ and $y = \cos x$ on the interval $\frac{\pi}{6} \leq x \leq \frac{\pi}{2}$.



25 Find the area between the two curves of $y = \sin 8x$ and $y = \cos 8x$ on the interval $\left[0, \frac{\pi}{16}\right]$.

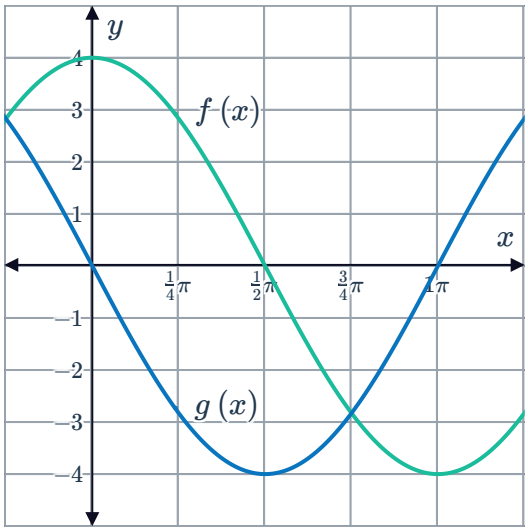


26 Find the area enclosed by the two curves of $f(x) = 5\cos 2x$ and $g(x) = 5\sin\left(\frac{x}{2}\right)$ between $x = \frac{\pi}{5}$ and $x = \pi$. Round your answer to two decimal places.



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27 Find the area of the region bounded by $f(x) = 4\cos x$, $g(x) = -4\cos\left(x - \frac{\pi}{2}\right)$, the y-axis and $x = \frac{3\pi}{4}$.



28 Find the area of the region bounded by the line $y = 2\pi - x$ and the curve $y = -2\sin x$ and the x-axis between $x = 0$ and $x = \frac{2\pi}{3}$.

29 Consider the functions $y = \sin x$ and $y = \cos x$.

- **a** Graph the functions on the same set of axes in the interval $[0, \pi]$.
- **b** Hence, find the area between the two curves in this interval.

30 Consider the functions $y = \sin 4x$ and $y = \cos 4x$ for $0 \leq x \leq \frac{\pi}{2}$.

- **a** Find the x-coordinates of the points of intersection of the two curves in the given interval.
- **b** Graph the two functions on the same set of axes.
- **c** Hence, find the area bounded by the two curves between the points of intersection found in part (a).

31 Consider the functions $f(x) = \sin 4x$ and $g(x) = 1 - \cos 4x$.

- **a** Graph the functions from 0 to $\frac{\pi}{2}$ on the same set of axes.
- **b** For what values of x is $f(x) = g(x)$?
- **c** Evaluate $\int_0^{\frac{\pi}{2}} (1 - \cos 4x - \sin 4x)\, dx$.
- **d** Find the area between the two curves on the interval $\left[0, \frac{\pi}{2}\right]$.

32 Find the exact area between the following graphs and lines:

- **a** $y = x^3 - 8$, $x = 2$, and $y = -8$
- **b** $y = e^x$, $y = e^{-x}$, and $x = 3$
- **c** $y = e^{5x}$, $y = e^{-5x}$, $x = -2$, and $x = 2$
- **d** $y = x$, $y = e^{1-x}$, and $x = 3$

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33 Consider the functions $f(x) = e^x + 12$ and $g(x) = 4x + e^3$.

a Find the x-coordinate of the point of intersection that is to the right of origin.

b Find the area in the first quadrant bounded by the two curves and the y-axis.

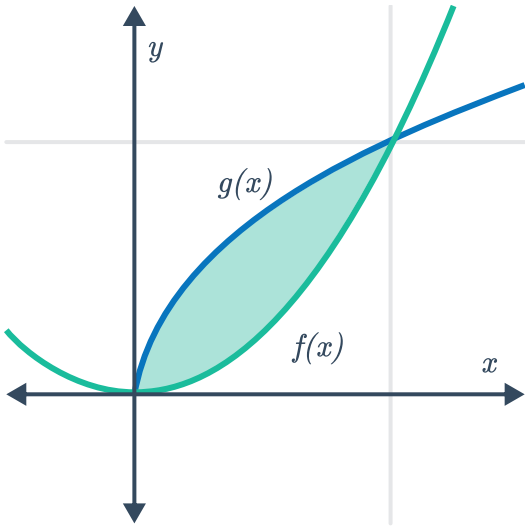
34 Consider the graphs of $f(x) = x^2$ and $g(x) = \sqrt{x}$:

a Find the x-coordinates of the points of intersection.

b Find the shaded area bounded by the two curves.

c Complete the following table for the area bounded by $f(x) = x^n$ and $g(x) = \sqrt[n]{x}$.

n	2	3	5	8
Area				



d Make a conjecture for the value of the area between the curves $f(x) = x^n$ and $g(x) = \sqrt[n]{x}$ for positive integer values of n.

e Prove your conjecture.

35 Consider the graph below which has the following properties:

- Functions $f(x)$ and $g(x)$ intersect at $x = 0$, $x = 2$ and $x = 3$
- Region A has an area of 22 square units
- Regions A and B have a combined area of 56 square units
- $\int_2^3 f(x)\,dx = -18$ and $\int_2^3 g(x)\,dx = -13$

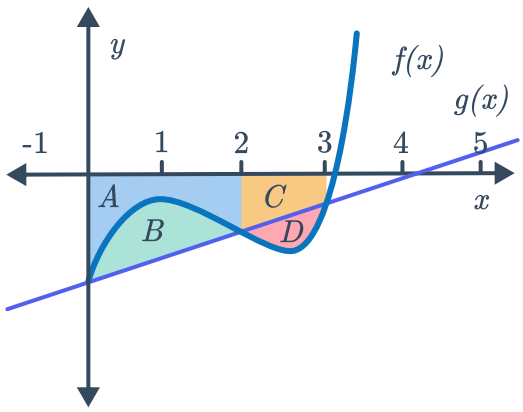
a Find $\int_0^2 f(x)\,dx$.

b Find $\int_0^2 (f(x) - g(x))\,dx$.

c Find the area of region C.

d Find the area of region D.

e Find $\int_0^3 (f(x) - g(x))\,dx$.



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36 Consider the graph below which has the following properties:

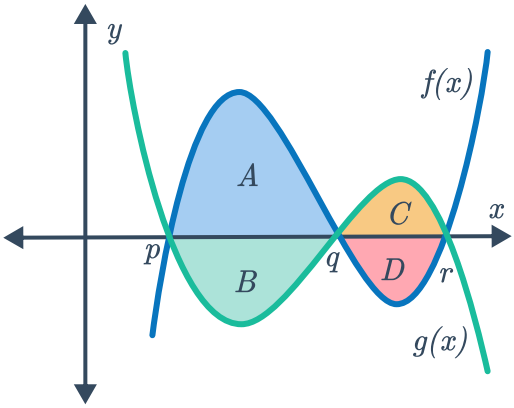
- Functions $f(x)$ and $g(x)$ intersect on the x-axis at points p, q, and r
- Region A has an area of 16 units^2
- Regions A and B have a combined area of 26 units^2
- $\int_p^r f(x)\,dx = 11$ and $\int_p^r g(x)\,dx = -9$

a Find the value of $\int_p^q f(x)\,dx$.

b Find the value of $\int_p^q g(x)\,dx$.

c Find the area of region C.

d Find the value of $\int_p^r (f(x) - g(x))\,dx$.



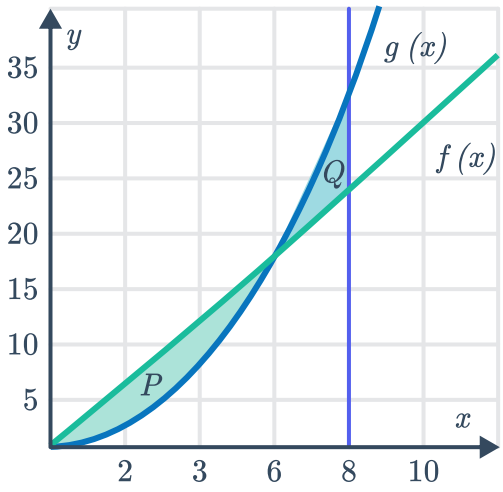
37 Consider the graph of the functions $f(x) = 3x$ and $g(x) = \dfrac{x^2}{2}$ together with the line $x = 8$:

Region P is the area enclosed by f and g.
Region Q is the area enclosed by f and g and $x = 8$.

a Find the area of region P.

b Find the area of region Q.

c $f(x)$ is redefined such that $f(x) = ax$ and the area of region P is half the area of region Q. Calculate the value of a that makes the statement true.



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